

Shrey Pandey

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EXPERIENCE

- **Coinbase** Bangalore, India
Machine Learning Engineer *Jun 2025 - Present*
 - **Multi-Agent System for Compliance Investigations:**
 - * Built a production multi-agent compliance investigation system using an orchestrator, specialized LLM agents, MCP servers, and compliance policy documents to decide regulatory escalation, detect risk typologies, and generate analyst-ready narratives.
 - * Owned the architecture and development of a five-part production stack including analyst frontend, gateway, queue-based engine, investigation agents, and MCP servers.
 - * Reached 94% accuracy on escalation decisions, 84% on typology detection, and 80% on overall narrative quality through analyst feedback loops, error analysis, and iterative prompt tuning.
- **Microsoft Research** Bangalore, India
Research Fellow *Jan 2024 - May 2025*
 - **Auto Azure SDK Migration:**
 - * Achieved 50% reduction in engineering effort for Azure SDK migration, saving 30 person-months annually, by developing an LLM-based framework to migrate Azure SDK code to the latest API version.
 - * Leveraged a hybrid approach combining symbolic code-graph analysis with LLM workflows and a refined knowledge base to streamline code generation.
 - **Programming with Representations [PwR] :**
 - * Designed PwR, an AI-assisted programming system enabling non-profits without engineering teams to build custom software by describing requirements in natural language.
 - * Developed an LLM-based pipeline transforming requirements into a custom DSL, compiled into Python code and rendered as a visual workflow chart for non-technical users. Open-sourced under JugalBandi and used as the foundation for its chatbot use cases.
 - **Jugalbandi [Docs] [Manager] [Studio]:**
 - * Led JugalBandi, an initiative built on PwR enabling non-tech organizations to develop and deploy AI-powered conversational bots. Featured in **Satya Nadella's** keynote talks.
 - * Led JugalBandi-Manager, a production runtime with a pluggable architecture for chat interfaces, LLM, vernacular, and voice/TTS providers, with FSM-based conversation flows.
 - * Deployed at CPGRAMS, MySchemes, and Bandhu (housing), with the open-source platform adopted by additional government and civil society initiatives.
 - **Dialog Systems for Government Services:**
 - * Designed LLM-based dialog systems for government agencies including the Ministry of Law and Justice, Prime Minister's Office, and NeGD, enhancing citizens' access to public services.
 - * Built a dense retrieval pipeline with LLM-driven follow-up question generation to classify citizen queries against a 14,000+ form grievance taxonomy and resolve ambiguity before filing.
- **Myntra (Walmart)** Bangalore, India
Senior Data Science Engineer, previously Data Science Engineer *Jul 2021 - Jan 2024*
 - **NNFetch - In-house Vector Database**
 - * Achieved **99%ile latency of 20ms at 1M req/min throughput** by developing an in-house vector database that unified embedding search across recommendation systems, personalized ranking, advertising, and semantic retrieval use cases.
 - * Implemented HNSW and ANNOY indexing with metadata filtering and high-frequency re-indexing to support real-time ML retrieval at scale.
 - **Ranking and Recommendation Systems**

- * Scaled personalized search and product recommendation systems to **5M req/min throughput** with **99%ile latency of 40ms**.
- * Built ranking and recommendation flows using user behavior signals and embeddings, with click-driven tuning, Multi-Criteria Decision Analysis (MCDA), and DPP-based diversification under business-aware optimization constraints.
- **ML Systems Optimization:** Developed reusable components to scale ML workflows.
 - * Delivered **15x higher throughput** and **88% annual hardware cost savings** across recommendation systems through a suite of reusable performance optimizations.
 - * Engineered a memory-efficient solution consolidating embeddings in OS shared memory, **reducing latency by 4.5x** and **memory footprint by 6x**.
 - * Implemented memory-layout optimization and increased register-level parallelism for similarity-score computation, reducing clock cycles and improving cache hit rates.
 - * Optimized DPP and MCDA computations in the post-retrieval ranking stack.

OPEN SOURCE PROJECTS

-  **Jugalbandi-Manager** [[Code](#)] ★38 🗨️43
Jugalbandi Manager is an AI-chatbot management platform enabling voice, multilingual and RAG capabilities

EDUCATION

- **Motilal Nehru National Institute of Technology** Allahabad, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.43/10.0 *July 2017 – June 2021*

PUBLICATIONS

- **[Best Industry Paper, Runner-Up] Rethinking ‘Complement’ Recommendations at Scale with SIMD** [[Paper](#)]
 Shrey Pandey, Saikat Kumar Das, Hrishikesh V. Ganu, and Satyaajeet Singh
15th ACM/SPEC International Conference on Performance Engineering [ICPE 24]
- **Efficient Vector Store System for Python using Shared Memory** [[Paper](#)]
 Dhruv Patel, Shrey Pandey, and Abhishek Sharma
2nd International Conference on AI-ML Systems [AI-MLSys 22]
- **CLOTON: A GAN based approach for Clothing Try-On** [[Paper](#)]
 Shrey Pandey, Yash Srivastava, Yukta Meena, and Rupesh Kumar Dewang
8th International Conference on Signal Processing and Integrated Networks [SPIN 21]

HONORS AND AWARDS

- **Best Industry Paper, Runner-Up** at ICPE 24 for ‘Rethinking Complement Product Recommendations at Scale Using SIMD’.
- **Employee of the Year 2023** amongst 500+ engineers at Myntra.
- **Best Long Paper Award** at Convergence 2023, Flipkart Data Science Conference.

SKILLS

- **LLM and Agent Systems:** Multi-agent systems, LLM applications, Retrieval-Augmented Generation (RAG), Agent evaluation, Prompt tuning, LangGraph, LangSmith, MCP
- **Languages:** Python, GoLang, C++
- **Machine Learning:** Recommendation systems, Information Retrieval, Ranking, Deep Learning, NLP
- **Frameworks and Tools:** PyTorch, ONNX, Triton, FastAPI, Flask, Kafka, Docker
- **Systems and Infrastructure:** Vector databases, HNSW, ANNOY, Metadata filtering, Distributed ML systems, Low-latency inference, Model serving, Shared-memory optimization